

Expert Behavioral Labeling Guide

SegPro-Med User Behavior Analytics

Purpose: This document provides the methodology for labeling experts based on their interaction patterns in the medical imaging annotation platform, using event stream, session-based, and hierarchical state transition data.

EXPERT PROFILES GENERATED FROM SAMPLE DATA

Expert 1: Dr. Smith

Session ID: sess_20251110_smith_001

Primary Dataset: Brain MRI (FLAIR)

Raw Metrics from Data:

- **Session Duration:** 12.5 minutes
- **Total Annotations:** 47 annotations
- **Slices Annotated:** 15 slices
- **AI Usage:** 26% (2 AI-accepted out of ~7 total annotations in AI stage)
- **Primary Tool:** drawrect (Rectangle drawing)
- **Avg Time/Annotation:** 15.9 seconds
- **Window Adjustments:** 1 (7% rate)
- **View Changes:** 2 (13% rate)
- **Edit Rate:** 0% (no edits)
- **AI Acceptance Rate:** 67% (2/3 suggestions)
- **Efficiency Score:** 0.89
- **Consistency Score:** 0.92

Applied Labels:

1. **Hybrid User** (AI Assistance Rate = 26%)
 - *Rationale:* Below 30% threshold, borderline between Manual-First and Hybrid. Uses AI selectively (only 1 invocation) but accepts majority of suggestions.
2. **Rectangle Specialist**
 - *Rationale:* Exclusive use of drawrect tool (100% usage in manual annotation phase).
3. **Speed Demon** (15.9 sec/annotation)
 - *Rationale:* Well below 45 sec threshold. Very rapid annotation pace.
4. **Expert** (Experience Level)
 - *Rationale:*
 - Zero edit rate (highly confident initial annotations)
 - High efficiency score (0.89)

- High consistency score (0.92)
- Streamlined workflow with minimal tool switching
- 5. **Brain MRI Specialist**
 - *Rationale:* Working on brain dataset with FLAIR modality, using appropriate labels (Tumor, Edema, Necrosis).
- 6. **Oncology Focused**
 - *Rationale:* Primary labels are Tumor, Edema, Necrosis - all cancer-related pathologies.
- 7. **Sequential Navigator**
 - *Rationale:*
 - Navigation pattern indicates linear progression (slice 45→46→47)
 - Low view switching rate (13%)
 - Uses “next” button for sequential navigation
- 8. **One-Pass Annotator**
 - *Rationale:* Zero edit rate suggests confident initial annotations without revision.
- 9. **Default Settings User**
 - *Rationale:* Only 1 window adjustment across entire session (7% rate).
- 10. **VLM Non-User**
 - *Rationale:* No evidence of MedGemma, Med-R1, or SmolVLM usage in session.
- 11. **High Efficiency** (Efficiency Score = 0.89)
 - *Rationale:* Above 0.8 threshold with rapid annotation pace and minimal errors.
- 12. **Highly Consistent** (Consistency Score = 0.92)
 - *Rationale:* Above 0.85 threshold indicating reproducible annotation patterns.

Final Persona: “The Efficient Brain AI-Hybrid” Summary: Expert-level brain MRI specialist who uses rectangle tools exclusively for rapid tumor annotation, selectively leverages AI when beneficial, maintains high quality through confident one-pass annotations, and prioritizes speed without sacrificing consistency.

Actionable Insights: - High productivity - suitable for time-sensitive campaigns - Consistent quality - reliable for gold standard creation - Could benefit from VLM training for label suggestions - May need periodic quality audits due to speed (no self-review) - Good AI adoption - accepts majority of suggestions when used

Expert 2: Dr. Johnson

Session ID: sess_20251110_johnson_001

Primary Dataset: Mammography (MG)

Raw Metrics from Data:

- **Session Duration:** 14.7 minutes
- **Total Annotations:** 8 annotations
- **Views Annotated:** 4 (LCC, RCC, LMLO, RMLO)
- **AI Usage:** 0%
- **Primary Tool:** `drawclosedpath` (Polygon/freehand)
- **Avg Time/Annotation:** 100.9 seconds
- **Window Adjustments:** 1 (12.5% rate)
- **View Changes:** 3 (33% rate)
- **Edit Rate:** 37.5% (3 edits out of 8 annotations)
- **Efficiency Score:** 0.72
- **Consistency Score:** 0.88

Applied Labels:

1. **Manual-First User** (AI Assistance Rate = 0%)
 - *Rationale:* Zero AI usage throughout entire session.
2. **Freehand Annotator**
 - *Rationale:* Exclusive use of `drawclosedpath` for complex polygon annotations (100% usage).
3. **Meticulous Reviewer** (100.9 sec/annotation)
 - *Rationale:* Well above 120 sec threshold when factoring in 87 sec creation + 93 sec edit time.
4. **Expert** (Experience Level)
 - *Rationale:*
 - High consistency score (0.88)
 - Systematic multi-view approach
 - Domain specialization in mammography
 - Confidence in tool selection
5. **Mammography Expert**
 - *Rationale:*
 - Works exclusively on MG modality
 - Uses all standard mammography views (LCC, RCC, LMLO, RMLO)
 - Appropriate labels (Mass, Calcification, Architectural Distortion)
6. **Mixed Pathology Annotator**
 - *Rationale:* Uses 3 diverse labels covering different breast findings (mass, calcifications, architectural changes).
7. **View Switcher**
 - *Rationale:*
 - 33% view switching rate (high)
 - Systematic coverage of all 4 mammography views
 - Multi-planar review approach
8. **Edit-Heavy Reviewer**

- *Rationale:* 37.5% edit rate (above 30% threshold) indicating iterative refinement.
9. **Default Settings User**
 - *Rationale:* Only 1 window adjustment (12.5% rate), relies on default visualization.
 10. **VLM Non-User**
 - *Rationale:* No VLM engagement detected.
 11. **Medium Efficiency** (Efficiency Score = 0.72)
 - *Rationale:* Between 0.6-0.8 range, balanced pace with quality focus.
 12. **Moderately Consistent** (Consistency Score = 0.88)
 - *Rationale:* In 0.7-0.85 range (note: actually on high end, could arguably be “Highly Consistent”).

Final Persona: “The Meticulous Mammography Craftsperson”

Summary: Expert mammography specialist who relies entirely on manual freehand annotation for precise delineation of breast lesions, emphasizes quality through iterative refinement across all standard views, and maintains systematic multi-planar review despite slower throughput.

Actionable Insights: - Exceptional quality focus - ideal for complex mammography cases - Comprehensive multi-view assessment ensures no missed findings - Lower productivity - not suitable for high-volume campaigns - Could benefit from AI assistance to improve efficiency - May need window optimization training to reduce reliance on defaults - High expertise in domain - good candidate for mentorship/training roles

Expert 3: Dr. Lee

Session ID: sess_20251110_lee_001

Primary Dataset: Abdomen CT

Raw Metrics from Data:

- **Session Duration:** 75.5 minutes
- **Total Annotations:** 62 annotations
- **Slices Annotated:** 28 slices
- **AI Usage:** 64.5% (20/62 annotations AI-assisted)
- **Primary Tool:** drawrect
- **Avg Time/Annotation:** 72.9 seconds
- **Window Adjustments:** 8 (13% rate)
- **View Changes:** 20 (27% rate - combining exploration + review phases)
- **Edit Rate:** 11.4% (5 edits)
- **Deletion Rate:** 3.2% (2 deletions)
- **AI Acceptance Rate:** 83.3% (20/24 suggestions)
- **VLM Usage:** Yes (MedGemma, 3 prompts, voice input)

- **Efficiency Score:** 0.81
- **Consistency Score:** 0.85

Applied Labels:

1. **AI-Heavy User** (AI Assistance Rate = 64.5%)
 - *Rationale:* Above 60% threshold with heavy reliance on MedSAM2.
2. **Rectangle Specialist**
 - *Rationale:* Primary tool is **drawrect** for organ and lesion bounding boxes.
3. **Balanced Annotator** (72.9 sec/annotation)
 - *Rationale:* Within 45-120 sec range, moderate pace.
4. **Intermediate** (Experience Level)
 - *Rationale:*
 - Moderate edit rate (11.4%, in 0.2-0.4 range)
 - Uses AI heavily but still requires manual refinement
 - Some deletions (3.2%) indicating experimental approach
 - Good but not exceptional efficiency (0.81)
5. **Abdominal Imaging Specialist**
 - *Rationale:*
 - Works on abdomen dataset
 - Multi-organ annotation (Liver, Kidney, Spleen)
 - Lesion/tumor detection expertise
6. **Mixed Pathology Annotator**
 - *Rationale:* Uses 5 diverse labels covering organs + pathology (Liver, Kidney, Spleen, Lesion, Tumor).
7. **Jump Scanner**
 - *Rationale:*
 - Exploration phase shows 45 slices viewed in 270 seconds
 - Non-sequential navigation to target specific regions
 - Efficient for known pathology
8. **View Switcher**
 - *Rationale:*
 - 20 total view changes (12 in exploration + 8 in review)
 - 27% view switching rate
 - Multi-planar review approach
9. **Edit-Heavy Reviewer**
 - *Rationale:* Combines with edit rate of 11.4% (borderline, but with deletions pushes toward refinement focus).
10. **Window Tweaker**
 - *Rationale:* 8 window adjustments across session (13% rate), actively optimizes visualization.
11. **VLM Power User**
 - *Rationale:*
 - Uses MedGemma for analysis
 - 3 custom prompts submitted

- Voice-to-text input utilized
- Leverages label suggestions (8/12 accepted)
- 12. **High Efficiency** (Efficiency Score = 0.81)
 - *Rationale:* Above 0.8 threshold despite longer session.
- 13. **Moderately Consistent** (Consistency Score = 0.85)
 - *Rationale:* Upper end of 0.7-0.85 range (could argue Highly Consistent).

Final Persona: “The AI-Powered Multi-Organ Analyst” Summary:
 Intermediate-level abdominal imaging specialist who heavily leverages AI and VLM tools for efficient multi-organ annotation, demonstrates strategic navigation with jump scanning, maintains quality through iterative refinement and multi-view review, and actively optimizes image visualization.

Actionable Insights: - Strong AI adoption - excellent candidate for AI workflow optimization - VLM engagement shows innovation and learning mindset - Good balance of speed and quality - Deletion rate (3.2%) suggests some uncertainty - may benefit from decision support guidelines - Could reduce edit rate with better AI model tuning or training - Multi-modal expertise developing - suitable for diverse case assignments - Window optimization habits show attention to detail

LABELING METHODOLOGY

Step 1: Extract Quantitative Metrics

From event stream, session, or state transition data, calculate:

```
# Example calculations
ai_assistance_rate = ai_annotations / total_annotations
avg_time_per_annotation = total_annotation_time / total_annotations
edit_rate = annotations_edited / total_annotations
deletion_rate = annotations_deleted / total_annotations
view_switching_rate = view_changes / total_events
window_adjustment_rate = window_adjustments / total_events
ai_acceptance_rate = ai_accepted / ai_suggestions_received
```

Step 2: Apply Threshold-Based Rules

Match metrics to categorical thresholds defined in `BEHAVIORAL_CATEGORIES.md`:

Category	Metric	Threshold
AI-Heavy User	ai_assistance_rate	> 0.60
Hybrid User	ai_assistance_rate	0.30 - 0.60
Manual-First User	ai_assistance_rate	< 0.30

Category	Metric	Threshold
Speed Demon	avg_time_per_annotation	< 45 sec
Balanced Annotator	avg_time_per_annotation	45-120 sec
Meticulous Reviewer	avg_time_per_annotation	> 120 sec
Edit-Heavy Reviewer	edit_rate	> 0.30
One-Pass Annotator	edit_rate	< 0.10
High Efficiency	efficiency_score	> 0.80
Highly Consistent	consistency_score	> 0.85

Step 3: Infer Behavioral Patterns

From state transitions and workflow sequences:

Sequential Navigator: - Look for linear slice progression ($n \rightarrow n+1 \rightarrow n+2$)
- Low jump frequency - State transitions: `slice_exploration` $\rightarrow$ `annotation` $\rightarrow$ `slice_exploration` (repeat)

Jump Scanner: - Frequent non-sequential slice changes - State pattern: `slice_exploration` $\rightarrow$ `jump` $\rightarrow$ `targeted_annotation`

View Switcher: - Frequent `view_changed` events - State transitions between different anatomical orientations - Multi-view review states in hierarchy

Step 4: Domain Specialization Detection

From metadata and label usage:

Dataset Analysis:

```
if dataset == "brain" and modality in ["flair", "t1", "t1c", "t2"]:
    labels.append("Brain MRI Specialist")
elif dataset == "MG" and modality == "MG":
    labels.append("Mammography Expert")
elif dataset == "abdomen":
    labels.append("Abdominal Imaging Specialist")
```

Pathology Focus:

```
oncology_labels = ["Tumor", "Necrosis", "Edema", "Mass", "Lesion"]
structural_labels = ["Organ", "ROI", "Normal Tissue", "Liver", "Kidney"]

if oncology_label_ratio > 0.6:
    labels.append("Oncology Focused")
elif structural_label_ratio > 0.6:
    labels.append("Structural/Anatomical Focused")
else:
    labels.append("Mixed Pathology Annotator")
```

Step 5: Tool and Interaction Analysis

```
# Tool preference
tool_usage = count_tool_usage(events)
if tool_usage["drawrect"] > 0.6:
    labels.append("Rectangle Specialist")
elif tool_usage["drawclosedpath"] + tool_usage["drawopenpath"] > 0.5:
    labels.append("Freehand Annotator")
elif len(tool_usage) >= 4:
    labels.append("Multi-Tool Expert")

# VLM usage
if "MedGemma" in events or "Med-R1" in events or "SmolVLM" in events:
    vlm_frequency = count_vlm_usage(events)
    if vlm_frequency > 0.2:
        labels.append("VLM Power User")
    elif vlm_frequency > 0.05:
        labels.append("VLM Occasional User")
else:
    labels.append("VLM Non-User")
```

Step 6: Experience Level Inference

Multi-factor assessment:

```
experience_score = 0
```

```
# Positive indicators
if consistency_score > 0.85: experience_score += 2
if edit_rate < 0.2: experience_score += 2
if efficiency_score > 0.8: experience_score += 1
if deletion_rate < 0.05: experience_score += 1
if tool_diversity <= 2: experience_score += 1 # Confident tool selection

# Negative indicators
if edit_rate > 0.4: experience_score -= 2
if deletion_rate > 0.15: experience_score -= 2
if inconsistent_workflow: experience_score -= 1

# Classification
if experience_score >= 5: label = "Expert"
elif experience_score >= 2: label = "Intermediate"
else: label = "Novice"
```

VALIDATION CHECKS

Consistency Validation

Ensure labels don't contradict:

```
# Contradictory pairs to flag
contradictions = [
    ("AI-Heavy User", "Manual-First User"),
    ("Speed Demon", "Meticulous Reviewer"),
    ("One-Pass Annotator", "Edit-Heavy Reviewer"),
    ("Expert", "Novice")
]

for label_pair in contradictions:
    if label_pair[0] in labels and label_pair[1] in labels:
        raise ValueError(f"Contradictory labels: {label_pair}")
```

Completeness Validation

Ensure key dimensions are covered:

```
required_dimensions = [
    "annotation_approach", # AI dependency
    "tool_preference",
    "speed_profile",
    "expertise_level",
    "domain_specialization"
]

for dimension in required_dimensions:
    if not any(label in dimension_labels[dimension] for label in labels):
        warnings.warn(f"Missing {dimension} label")
```

EXAMPLE LABELING ALGORITHM (Python Pseudocode)

```
def label_expert(session_data):
    labels = []

    # Extract metrics
    metrics = calculate_metrics(session_data)

    # AI Dependency
    if metrics['ai_rate'] > 0.6:
        labels.append("AI-Heavy User")
```

```

elif metrics['ai_rate'] > 0.3:
    labels.append("Hybrid User")
else:
    labels.append("Manual-First User")

# Tool Preference
primary_tool = metrics['primary_tool']
if primary_tool == 'drawrect':
    labels.append("Rectangle Specialist")
elif primary_tool in ['drawclosedpath', 'drawopenpath']:
    labels.append("Freehand Annotator")

# Speed Profile
if metrics['avg_time_per_annotation'] < 45:
    labels.append("Speed Demon")
elif metrics['avg_time_per_annotation'] < 120:
    labels.append("Balanced Annotator")
else:
    labels.append("Meticulous Reviewer")

# Experience Level
exp_score = calculate_experience_score(metrics)
if exp_score >= 5:
    labels.append("Expert")
elif exp_score >= 2:
    labels.append("Intermediate")
else:
    labels.append("Novice")

# Domain Specialization
if session_data['dataset'] == 'brain':
    labels.append("Brain MRI Specialist")
elif session_data['dataset'] == 'MG':
    labels.append("Mammography Expert")
elif session_data['dataset'] == 'abdomen':
    labels.append("Abdominal Imaging Specialist")

# Navigation Style
if metrics['sequential_navigation_rate'] > 0.7:
    labels.append("Sequential Navigator")
elif metrics['jump_navigation_rate'] > 0.4:
    labels.append("Jump Scanner")

if metrics['view_switching_rate'] > 0.1:
    labels.append("View Switcher")

```

```

# Review Behavior
if metrics['edit_rate'] > 0.3:
    labels.append("Edit-Heavy Reviewer")
elif metrics['edit_rate'] < 0.1:
    labels.append("One-Pass Annotator")

# VLM Usage
if metrics['vlm_usage_rate'] > 0.2:
    labels.append("VLM Power User")
elif metrics['vlm_usage_rate'] > 0.05:
    labels.append("VLM Occasional User")
elif metrics['vlm_usage_rate'] == 0:
    labels.append("VLM Non-User")

# Efficiency & Consistency
if metrics['efficiency_score'] > 0.8:
    labels.append("High Efficiency")
elif metrics['efficiency_score'] > 0.6:
    labels.append("Medium Efficiency")
else:
    labels.append("Low Efficiency")

if metrics['consistency_score'] > 0.85:
    labels.append("Highly Consistent")
elif metrics['consistency_score'] > 0.7:
    labels.append("Moderately Consistent")
else:
    labels.append("Inconsistent")

# Validate and return
validate_labels(labels)
return labels

```

USAGE EXAMPLES

Example 1: Labeling from Event Stream

```

# Load event stream data
events = load_json("sample_event_stream_expert_dr_smith.json")

# Calculate metrics
total_annotations = count_events(events, event_type="annotation_created")
ai_annotations = count_events(events, metadata__ai_assistance=True)
ai_rate = ai_annotations / total_annotations

```

```

# Apply labels
labels = label_expert({
    'ai_rate': ai_rate,
    'primary_tool': most_common_tool(events),
    'dataset': events[0]['metadata']['dataset'],
    # ... other metrics
})

print(f"Expert Labels: {labels}")

```

Example 2: Generating Persona

```

labels = label_expert(session_data)
persona = generate_persona(labels)

print(f"Persona: {persona['name']}")
print(f"Summary: {persona['summary']}")
print(f"Insights: {persona['insights']}")

```

OUTPUT FORMAT

Labeled Expert JSON

```

{
  "user_id": "expert_dr_smith",
  "session_id": "sess_20251110_smith_001",
  "labels": [
    "Hybrid User",
    "Rectangle Specialist",
    "Speed Demon",
    "Expert",
    "Brain MRI Specialist",
    "Oncology Focused",
    "Sequential Navigator",
    "One-Pass Annotator",
    "VLM Non-User",
    "High Efficiency",
    "Highly Consistent"
  ],
  "persona": "The Efficient Brain AI-Hybrid",
  "metrics": {
    "ai_assistance_rate": 0.26,
    "avg_time_per_annotation": 15.9,
    "efficiency_score": 0.89,
    "consistency_score": 0.92
  }
}

```

```
},  
  "insights": [  
    "High productivity - suitable for time-sensitive campaigns",  
    "Consistent quality - reliable for gold standard creation",  
    "Could benefit from VLM training",  
    "May need periodic quality audits due to speed"  
  ]  
}
```

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